

Diagnosing Visual Similarity in Design Patent Citation Networks with Multimodal LLM Judgments

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Abstract

Prior-art search for design patents depends on visual comparison because design patents typically contain little text beyond a short claim and figure captions. Although embedding models can retrieve visually related drawings at scale, it remains unclear how well embedding similarity aligns with examiner-style visual assessment or where it fails to capture ornamental similarity. This paper uses USPTO examiner citations as a network lens for diagnosing zero-shot multimodal large language model (MLLM) judgments of design-patent similarity. Using IMPACT as the source corpus, we construct within-class examiner-citation pairs, obtain MLLM judgments of ornamental similarity, and compare these judgments with cosine similarity in a shared image-embedding space. We then form citation triplets, including citation chains ($A \rightarrow B \rightarrow C$) and co-citation structures ($A \rightarrow B \leftarrow C$), and introduce a residual measure that compares embedding similarity across adjacent citation edges. We use the zero-crossing point of the mean residual as a diagnostic boundary for identifying triplets that merit closer visual assessment. Beyond adjacent edges, we find that visual coherence in citation triplets is largely confined to same-applicant design families and does not extend across applicants, indicating that citation-neighborhood expansion mainly surfaces near-duplicate refilings rather than cross-applicant prior art.

1 Introduction

Design patents protect the ornamental appearance of products, including shape, contour, and overall visual impression, rather than their functional features (Samuelson and Gergen, 2019). Unlike utility patents, design patents usually contain little text beyond a short claim and figure captions (Shomee et al., 2024). Prior-art search for design patents therefore depends heavily on visual comparison: a relevant reference may be missed if search

relies only on textual metadata or broad design-class labels. This setting raises a practical question for computational search systems: how can large collections of design-patent drawings be screened efficiently while still preserving an interpretable account of visual similarity?

Two lines of work are relevant but incomplete for this purpose. First, embedding-based retrieval models can rank visually related patent drawings at scale, and recent design-patent datasets such as IMPACT and DeepPatent have made such analysis feasible (Shomee et al., 2024; Kucer et al., 2022). However, embedding similarity alone does not explain whether a retrieved pair shares examiner-relevant ornamental features. Second, patent citation networks have been widely used to study relational structure in utility patents (Chakraborty et al., 2020), and motif analysis shows that small directed subgraphs can provide interpretable local patterns in complex networks (Ohnishi et al., 2010). Yet design-patent citation networks remain under-explored, even though examiner citations provide a natural source of prior-art relationships grounded in visual comparison.

This paper studies examiner-cited design patents as a visual citation network. Using IMPACT as the source corpus, we obtain USPTO examiner-citation pairs for design patents and restrict the main analysis to within-class citations. For each citation pair, we compare two complementary signals. The first is a zero-shot MLLM judgment, produced by prompting a multimodal model to decide whether two patent drawings convey a substantially similar ornamental impression. The second is cosine similarity between image embeddings computed from a shared multimodal embedding space. We then move beyond isolated citation pairs by constructing triplet-level citation motifs, including citation chains ($A \rightarrow B \rightarrow C$) and co-citation structures ($A \rightarrow B \leftarrow C$). These triplets allow us to ask whether visual similarity is locally consistent

across neighboring examiner citations.

For these citation structures, we introduce a residual measure that compares embedding similarity between adjacent citation edges in citation-chain and co-citation triplets. This residual allows us to test whether the embedding space orders neighboring examiner citations in a way that is consistent with MLLM-based visual similarity judgments. We further use the zero-crossing point of the mean residual as a diagnostic boundary for identifying triplets that require closer visual similarity assessment.

2 Related Work

Large-scale design-patent datasets have enabled computational work on design retrieval. Shomee et al. (2024) introduced IMPACT, a multimodal corpus of U.S. design patents with drawings and metadata, and Kucer et al. (2022) introduced Deep-Patent for patent drawing recognition and retrieval. We use IMPACT as the source corpus for patent images, metadata, class labels, and identifiers. Unlike these dataset papers, our study examines examiner-cited design patents as a citation network and relates those citations to MLLM judgments and embedding similarity.

Patent citation networks have been used to characterize relational structure in patent data (Chakraborty et al., 2020). More generally, network motif analysis studies recurring small subgraphs as interpretable local patterns in directed networks (Ohnishi et al., 2010). Most prior patent-citation work, however, focuses on utility patents, where citations are often interpreted as technological or knowledge relationships. Design-patent citations differ because the relevant prior-art relationship is primarily visual and ornamental. This paper therefore analyzes examiner-citation neighborhoods—especially citation chains and co-citations—as triplet-level motifs that connect local citation topology to visual similarity signals.

Prior design-patent retrieval methods mainly rely on learned visual or multimodal representations. PatentCLIP, for example, improves retrieval by training on IMPACT image-caption pairs (Shomee et al., 2024), and Lo et al. (2024) combine LLM-generated image descriptions with contrastive learning. Separately, Ikoma and Mitamura (2025) show that LLMs can support novelty assessment for utility-patent claims. These studies motivate our question but do not address zero-shot

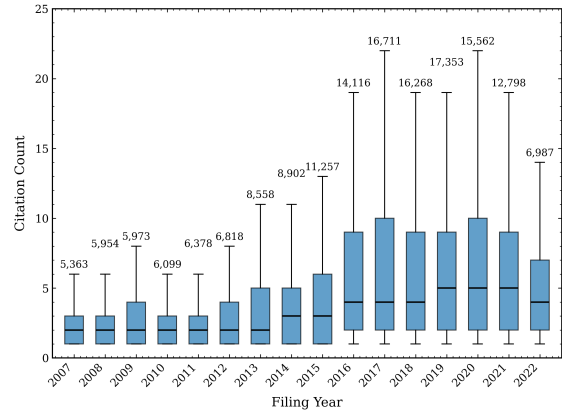


Figure 1: Distribution of citation counts by filing year for USPTO design patents (2007–2022). Box spans the interquartile range (Q1–Q3); the horizontal line indicates the median; whiskers extend to $1.5 \times \text{IQR}$. The number above each box is the total patent count for that year.

MLLM judgment of ornamental similarity from design-patent drawing pairs. We evaluate whether such judgments provide an interpretable signal that complements embedding-based similarity within the examiner-citation network.

3 Data and Method

This study draws on the IMPACT dataset (Shomee et al., 2024), which provides patent drawings and associated metadata for 434,497 U.S. design patents published between 2007 and 2022, together with 3,030,160 patent-drawing figures. Each record includes patent identifiers, titles, claims, publication dates, design-class labels, design-type annotations, image counts per patent, and viewpoint descriptions; drawings are stored as TIF images spanning multiple view types (perspective, front, and orthographic).

Using the patent identifiers in IMPACT, we query the USPTO API to obtain examiner-citation records and retain only pairs for which both patents are available in IMPACT. Records without an office-action date are excluded so that the resulting edges correspond to examiner-identified prior-art citations. This procedure yields 430,570 examiner-citation pairs: 385,617 within-class pairs (89.6%) and 44,953 cross-class pairs (10.4%). Because cross-class citations introduce heterogeneous visual contexts, all subsequent analyses are restricted to within-class pairs. Figure 1 shows the distribution of citation counts by filing year, and Appendix A.2 reports within- and cross-class cita-

tion counts for both the full reference set and the MLLM-judged similar subset.

For each within-class citation pair, we perform two complementary visual similarity assessments. First, we submit the image pair associated with the citation relationship to an MLLM (Qwen3-VL-4B-Instruct) using the prompt described in Appendix A.1. Inference is performed in bfloat16 on a single NVIDIA GeForce RTX 3090 GPU. The model returns a structured JSON object containing a binary similarity judgment (*Yes: similar / No: not similar*), a five-point confidence score, and a brief rationale based on the dominant visual features. We use these MLLM judgments to diagnose how perceived visual similarity varies with citation structure and design class. Second, for each design patent, we select one representative drawing—the perspective view when available and otherwise the front view—and encode it with a multimodal text–image embedding model (Qwen3-VL-Embedding-2B), producing a 2,048-dimensional vector for each image. Although the model computes these representations in 16-bit brain floating point (bfloat16), the final embedding vectors are output as 32-bit floating-point (FP32) arrays. For each image pair (A, B), we compute embedding similarity using cosine similarity:

$$\text{sim}(A, B) = \frac{\mathbf{v}_A^\top \mathbf{v}_B}{\|\mathbf{v}_A\|_2 \|\mathbf{v}_B\|_2}, \quad (1)$$

where $\mathbf{v}_A, \mathbf{v}_B \in \mathbb{R}^{2048}$ are the multimodal embeddings of images A and B , respectively.

In design-patent prior art search, pairwise similarity between a query design image A and a candidate image B is typically evaluated in isolation, leaving the reliability of MLLM-generated similarity signals undiagnosed with respect to the broader citation context. We address this by introducing a third design image C from the directed citation graph and using triplets of A, B , and C as the unit of reliability diagnosis. We define two triplet types:

Type 1 (citation chain, $A \rightarrow B \rightarrow C$)

A cites B and B cites C .

Type 2 (co-citation, $A \rightarrow B \leftarrow C$)

A and C independently cite B .

We exclude Type 3, the reversed citation chain ($A \leftarrow B \leftarrow C$), because it is the mirror image of Type 1 and does not introduce a distinct citation structure for the present analysis. For each type,

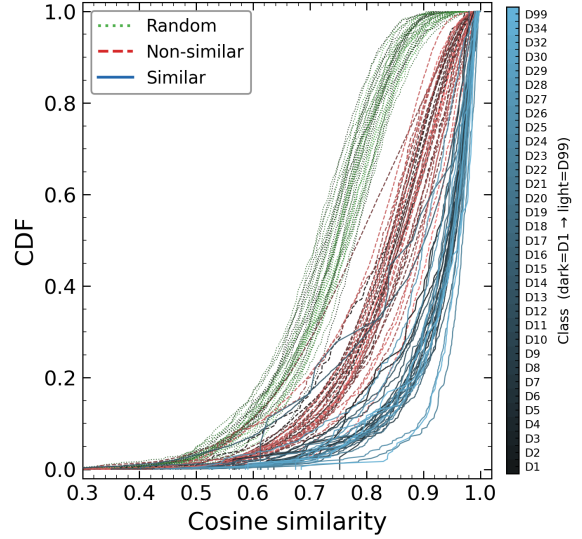


Figure 2: **Cosine similarity CDF for citations across all 33 design patent classes.** x axis, cosine similarity; y axis, empirical CDF. Four line types: solid blue, Similar (Yes); dashed red, Non-similar (No); solid gray, All judged; dotted dark gray, Random ($N = 1000$). Each line type contains 33 thin lines (one per class) colored from near-black (D1) to the standard type color (D99), ordered by class number (ascending); the right-hand colorbar shows the class–color correspondence. Similar pairs exhibit consistently higher cosine similarity than Non-similar or Random pairs across all classes.

we analyze the relationship between MLLM binary judgments (Yes/No) and embedding cosine similarity across constituent pairs, thereby diagnosing how triplet-level citation consistency shapes the reliability of individual similarity signals.

4 Results

We obtained cosine similarities and MLLM binary judgments for all valid citation pairs. For each design class listed in Table 3, we group citation pairs by their MLLM binary judgment and compare the corresponding cosine-similarity distributions. In most classes, pairs judged visually similar exhibit higher cosine similarity than pairs judged non-similar or randomly sampled pairs (Figure 2).

We next examine how citation pairs, MLLM binary judgments, and cosine similarities are related in local citation neighborhoods. We therefore move from isolated citation pairs to citation triplets to ask whether visual similarity propagates through the network. If citation networks form visually coherent chains, then the MLLM judgment and embedding similarity for one citation pair, $A \rightarrow B$, should help predict the visual similarity of the adja-

cent pair, $B \rightarrow C$. Conversely, if visual similarity is edge-specific, then a visually similar $A \rightarrow B$ pair need not imply that $B \rightarrow C$ is also visually similar.

We illustrate this procedure using Type 1 triplets as a representative case. Specifically, we stratify Type 1 triplets by the two MLLM judgments on $A \rightarrow B$ and $B \rightarrow C$. The four label regimes are

$$(x_{AB}, y_{BC}) \in \left\{ \begin{array}{l} (\text{Yes}, \text{Yes}), (\text{Yes}, \text{No}), \\ (\text{No}, \text{Yes}), (\text{No}, \text{No}) \end{array} \right\}. \quad (2)$$

For each triplet, we compute the residual

$$r = \text{sim}(B, C) - \text{sim}(A, B), \quad (3)$$

where $\text{sim}(\cdot)$ denotes cosine similarity. This residual measures whether the downstream edge is closer or farther in the embedding space than the upstream edge. We apply the same logic to Type 2 triplets by replacing $B \rightarrow C$ with $C \rightarrow B$; the full set of Type 1 and Type 2 variants is summarized in Figure 6.

Figure 4 isolates the Type 1 (Yes, No) regime, where “Yes” and “No” denote MLLM judgments rather than ground-truth similarity labels: $A \rightarrow B$ is judged visually similar by the MLLM, whereas $B \rightarrow C$ is judged non-similar. This setting is especially diagnostic because it tests how MLLM-assessed similarity behaves when an adjacent citation edge receives a conflicting MLLM judgment. Two findings emerge. First, the mean residual is positive when upstream embedding similarity is low ($r \approx +0.27$ at $\text{sim}(A, B) \approx 0.33$), decreases monotonically, crosses zero near $\text{sim}(A, B) \approx 0.78$, and becomes negative when upstream similarity is high ($r \approx -0.10$ at $\text{sim}(A, B) \approx 0.98$). This zero crossing separates two qualitatively different regimes. Below the threshold ($\text{sim}(A, B) \lesssim 0.78$), triplets in which the MLLM judgments $A \rightarrow B$ as visually similar tend to have a downstream edge with *higher* cosine similarity that is nonetheless labeled non-similar ($r > 0$). This inversion suggests that, in the lower-similarity regime, MLLM responses may reflect design-specific visual cues that are not fully represented by cosine proximity alone, although these responses should not be treated as ground truth. Above the threshold ($\text{sim}(A, B) \gtrsim 0.78$), the residual is negative and the direction of the MLLM judgments is consistent with cosine ordering: the upstream edge labeled similar has higher embedding similarity than the downstream edge labeled non-similar. In this

high-similarity regime, embedding proximity and MLLM judgment align more closely, so embedding distance serves as a more reliable proxy for the model’s visual verdict. Second, only 30.2% of triplets have a positive residual, suggesting that MLLM-assessed similarity for $A \rightarrow B$ rarely extends to $B \rightarrow C$ within this response regime. Taken together, these results suggest that similarity search by simply following design-patent citation chains may be unreliable without additional visual diagnosis. The downstream edge often appears to mark a break in MLLM-assessed visual consistency, and lower-similarity regions require additional judgment-based diagnosis.

The full set of judgment regimes in Figure 6 shows that this pattern is not specific to the Type 1 (Yes, No) case. Across both Type 1 citation chains and Type 2 co-citation structures, the residual is organized primarily by the relative MLLM judgments on the two edges. When both edges are judged similar, the residual is nearly symmetric around zero ($\bar{r} = +0.0004$ for Type 1 and $\bar{r} = +0.0113$ for Type 2), with positive residuals in 50.0% of Type 1 triplets and 55.8% of Type 2 triplets. When neither edge is judged similar, the much larger (No, No) regime likewise has a near-zero mean residual ($\bar{r} \approx +0.001$) and a positive fraction close to 50%, indicating little directional structure in the embedding space. By contrast, the asymmetric regimes move in opposite directions: (No, Yes) produces positive residuals for both Type 1 ($\bar{r} = +0.0518$, 69.8% positive) and Type 2 ($\bar{r} = +0.0567$, 72.2% positive), whereas (Yes, No) produces negative residuals for Type 1 ($\bar{r} = -0.0508$, 30.2% positive) and Type 2 ($\bar{r} = -0.0305$, 36.7% positive). Thus, the sign and magnitude of the residual track the relative visual judgment across adjacent edges rather than the triplet construction itself. This complements the local-consistency view: embedding similarity provides a continuous measure of proximity, but its interpretation as prior-art evidence becomes stronger when it agrees with the MLLM judgment pattern across neighboring citation edges.

5 Discussion

The triplet analysis shows that embedding similarity and MLLM judgments are related but not interchangeable signals. The residual $r = \text{sim}_{BC/CB} - \text{sim}_{AB}$ measures how the embedding space orders two adjacent citation edges, while the MLLM judgments indicate whether those

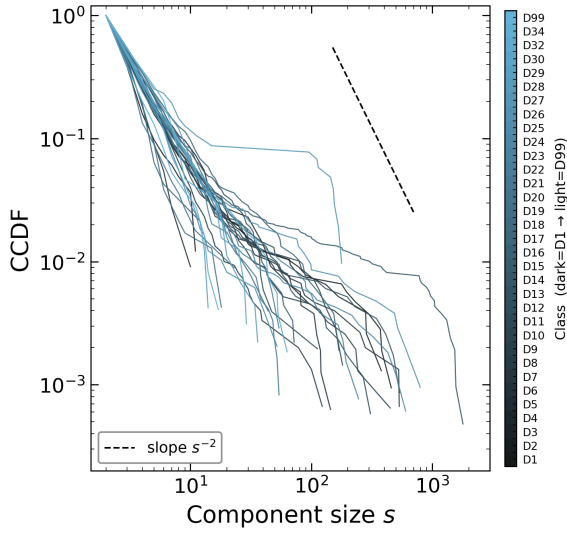


Figure 3: **Complementary cumulative distribution function (CCDF) of weakly connected component sizes for all 33 design patent classes.** Let S denote the size (number of nodes) of a weakly connected component drawn uniformly at random from a given class. The y axis shows the CCDF. The x axis shows component size s (log scale). Each thin line represents one class, colored from near-black (D1) to medium blue (D99) ordered by class number; the right-hand colorbar shows the class-color correspondence.

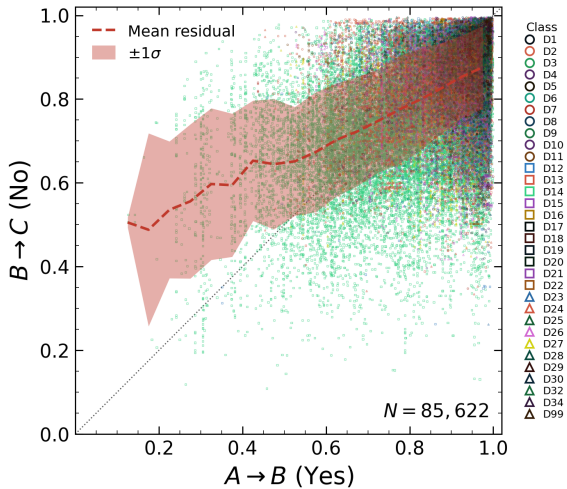


Figure 4: **Cosine similarity for $B \rightarrow C$ as a function of $A \rightarrow B$ ($A \rightarrow B$: Yes; $B \rightarrow C$: No).** Results are shown for $N = 85,622$ Type 1 triplets. The shaded band shows $\pm 1\sigma$ of the bin-wise residual $r = \text{sim}(B, C) - \text{sim}(A, B)$. Plot colors represent individual classes.

edges appear visually similar. Table 4 supports a simple interpretation: the residual is close to zero when the two edges receive the same MLLM judgment, but becomes signed when the judgments differ. As defined in the Results, the zero crossing is therefore not a general similarity threshold. In the (Yes, No) regime shown in Figure 4, for example, it marks the regime-specific boundary at which the embedding order changes from contradicting to supporting the MLLM ordering.

The (No, No) regimes require a different interpretation because the residual measures a relative ordering between two edges, not visual similarity itself. In these regimes, both $A \rightarrow B$ and the adjacent edge are judged visually non-similar by the MLLM. Their near-zero mean residuals (0.0009 for Type 1 and 0.0036 for Type 2) therefore indicate that, on average, the embedding model does not systematically place one of the two MLLM-judged No edges closer than the other. The positive-residual rates are also near chance, reinforcing that the comparison is not directionally biased.

This provides a useful check against a simple judgment-imbalance explanation. Although the (No, No) regimes dominate the triplet counts, their residuals do not show a systematic shift. The strongest residual signals instead appear in the asymmetric regimes, where the MLLM judgment changes across adjacent edges. This suggests that the residual captures a mismatch between embedding proximity and examiner-style visual similarity when the two signals disagree, rather than merely reflecting the overall prevalence of MLLM-judged No pairs.

A further implication is that Type 1 and Type 2 triplets diagnose different forms of local consistency. Type 1 triplets represent citation chains $A \rightarrow B \rightarrow C$ and therefore test whether visual similarity propagates along successive examiner citations. Their asymmetric regimes are nearly mirror images: (Yes, No) has 30.2% positive residuals, whereas (No, Yes) has 69.8%, with nearly equal sample sizes and similar absolute mean residuals. This symmetry suggests that, in citation chains, the residual mainly reflects the relative visual strength of the two edges rather than a privileged structural role. Type 2 triplets, by contrast, represent co-citation structures $A \rightarrow B \leftarrow C$, where B acts as a shared reference. Here the asymmetric regimes are less balanced: the (No, Yes) regime is larger and has a stronger positive mean residual than the corresponding negative residual in (Yes, No). This

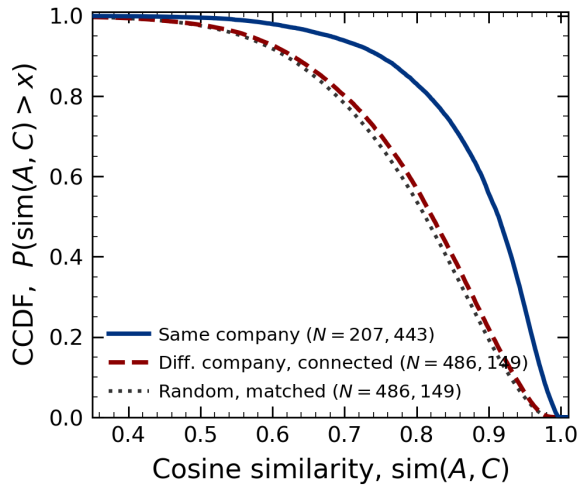


Figure 5: **Same-company A–C pairs strongly dominate connected different-company and random pairs in Type 2 co-citation triplets.** The CCDF shows $P(\text{sim}(A, C) > x)$ for same-company pairs (solid blue), different-company citation-connected pairs (dashed red), and class-distribution-matched random different-company pairs (dotted grey) in convergent triplets ($A \rightarrow B \leftarrow C$). The connected different-company and random curves nearly overlap, whereas the same-company curve remains substantially higher across the similarity range. This indicates that shared citation targets mainly expose visually coherent intra-firm design families rather than cross-company visual neighbors.

indicates that shared citation targets can behave as visual hubs that are selectively similar to some citing designs but not others. Co-citation evidence should therefore be treated as a hub-centered diagnostic, not as equivalent to a citation-chain signal.

Figure 5 clarifies what Type 2 co-citation neighborhoods contribute beyond individual citation edges. In these convergent triplets, two citing patents A and C share the same cited reference B , so a strong A – C similarity signal would indicate that co-citation can expand a query toward additional visually related designs. The observed pattern is more limited: same-company A – C pairs are consistently more similar, but connected different-company pairs are nearly indistinguishable from class-matched random pairs. Thus, for Type 2 triplets, visual coherence is driven primarily by same-applicant design families rather than by co-citation structure alone.

This finding narrows the role of the proposed framework. Embedding similarity and zero-shot MLLM judgment remain useful for triage: embeddings screen large candidate sets, MLLM judgments provide an interpretable edge-level assess-

ment, and the triplet residual highlights neighborhoods where these signals disagree. However, the Type 2 result shows that expanding through shared citation targets should not be treated as a general mechanism for discovering cross-applicant visual prior art. Instead, it is most useful for flagging near-duplicate refilings or visually coherent intra-firm design families. Validating the MLLM judgments against expert labels and evaluating end-to-end retrieval ($\text{recall}@k$, mean average precision) are the natural next steps.

Limitations

This study has several limitations. First, the analysis is restricted to USPTO design patents contained in IMPACT and to examiner-citation pairs for which both endpoints are available in that dataset. We further focus on within-class citations because cross-class citations are comparatively rare and visually heterogeneous. This choice makes the analysis more controlled, but it also limits the extent to which the findings generalize to cross-class prior-art search, other patent offices, or design domains that are underrepresented in IMPACT.

Second, the MLLM judgments should not be interpreted as legal determinations of design-patent similarity. The model is prompted to approximate an examiner-oriented visual comparison, but its binary Yes/No outputs are not a substitute for expert examination, legal standards, or human annotation. Model responses may also vary with prompt wording, image preprocessing, view selection, and model version. Future work should therefore compare multiple MLLMs, evaluate prompt robustness, and validate the judgments against expert labels or office-action outcomes where available.

Third, the embedding analysis uses a single representative drawing for each patent, preferring the perspective view when available and otherwise the front view. This simplifies full-corpus retrieval and triplet construction, but design patents often include multiple views that jointly define the claimed ornamental appearance. A multi-view representation may capture shape, contour, and surface features more completely than the single-view setting used here.

Fourth, the residual analysis is diagnostic rather than causal. A signed residual indicates how the embedding space orders two adjacent citation edges relative to the MLLM judgments, but it does not prove why the model judged one edge similar

and the other non-similar. The zero-crossing values are regime-specific summaries, not universal similarity thresholds. Validating MLLM judgments against expert labels and evaluating end-to-end retrieval (recall@*k*, mean average precision) are the natural next steps.

Acknowledgments

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A Example Appendix

A.1 Prompts

The box below presents the prompt used for visual similarity assessment between two design patent images. The prompt instructs the model to determine whether two cited design patent images convey a substantially similar overall visual impression from the perspective of a USPTO design patent examiner.

Prompt: Design Similarity Assessment

Act as an expert AI in intellectual property law. You are given Image A and Image B. Evaluate whether these two designs are similar under

the following unified US/EU legal standard: Would an observant buyer who is familiar with prior art consider the overall visual impression of the two designs to be substantially the same? Focus on: overall shape and form, surface ornamentation, and the combination of visual elements as a whole. Ignore non-visual features. Respond with ONLY a valid JSON object – no markdown, no code fences, no explanatory text before or after. Required JSON schema (use exactly these keys):

```
{
  "similarity": "Yes" or "No",
  "confidence": integer from 1 (very uncertain) to 5 (highly certain),
  "reason": "1-2 sentence rationale citing the dominant visual features that drove the decision"
}
```

A.2 Class-Level Citation Counts

A.3 Design Patent Classes

A.4 Triplet Analysis

Table 1: **Within-class citations overwhelmingly outnumber cross-class citations; the MLLM judges roughly 7–8% of within-class cited pairs as visually similar; and D14 accounts for over half of all triplets.** Within-class and cross-class references are examiner-cited patent pairs whose source and target patents share or differ in USPTO design class, respectively. MLLM-similar: subset judged visually similar (exact-match $\cup$ high-similar $\cup$ similar). Cross-class counts use the citing patent (source) as the class anchor. Triplet counts: ordered citation chains of length 3 used for cosine-similarity scatter-plot analysis.

Design class	Reference within	Reference cross	MLLM-similar within	MLLM-similar cross	Triplets Type 1	Triplets Type 2	Triplets Type 3
D01	400	27	57	1	315	301	315
D02	16,725	239	1,462	17	65,663	61,988	65,663
D03	8,446	2,688	499	78	20,650	19,223	20,650
D04	2,068	231	110	4	5,282	5,529	5,282
D05	379	44	48	0	166	158	166
D06	14,208	1,596	1,408	28	24,310	23,415	24,310
D07	17,352	2,486	1,563	47	47,718	46,227	47,718
D08	9,263	1,645	696	14	11,023	10,741	11,023
D09	13,324	2,596	1,346	58	14,835	13,849	14,835
D10	5,882	2,902	498	58	8,120	8,053	8,120
D11	3,846	714	415	7	6,727	6,230	6,727
D12	21,345	993	1,436	15	54,501	52,349	54,501
D13	16,976	5,276	1,138	81	49,192	46,816	49,192
D14	136,896	9,212	8,003	129	862,117	798,860	862,117
D15	9,293	1,104	808	3	24,175	22,816	24,175
D16	6,753	1,093	479	26	11,337	10,896	11,337
D17	176	34	22	1	146	164	146
D18	1,875	376	310	6	2,164	2,255	2,164
D19	926	182	112	7	490	448	490
D20	456	268	80	7	413	445	413
D21	10,521	1,579	1,197	29	26,637	24,885	26,637
D22	2,925	224	278	5	12,366	12,585	12,366
D23	13,674	1,451	1,583	19	29,451	27,145	29,451
D24	20,237	3,344	1,424	21	54,010	51,442	54,010
D25	3,067	514	594	5	5,485	5,344	5,485
D26	26,584	1,220	1,004	11	205,511	194,370	205,511
D27	8,985	820	310	7	73,373	71,518	73,373
D28	3,254	645	363	6	5,258	5,104	5,258
D29	1,149	384	91	1	813	771	813
D30	1,968	404	121	3	2,321	2,181	2,321
D32	5,035	505	492	2	12,021	11,454	12,021
D34	1,129	57	203	4	904	890	904
D99	500	100	84	0	752	724	752
Total	385,617	44,953	28,234	700	1,638,246	1,539,176	1,638,246

Table 2: **Descriptive statistics (33 USPTO design classes) for the citation and triplet columns of Table 1.** MLLM-non-similar counts are derived as (examiner-cited pairs within class) – (MLLM-similar pairs within class). Argmin/argmax class labels are given in the companion CSV.

Column	Sum	Mean	Median	Std	Min	Max
Examiner-cited pairs, within class	385,617	11,685.4	5,882.0	23,612.4	176	136,896
Examiner-cited pairs, cross class	44,953	1,362.2	714.0	1,843.9	27	9,212
MLLM-similar pairs, within class	28,234	855.6	492.0	1,389.0	22	8,003
MLLM-similar pairs, cross class	700	21.2	7.0	29.4	0	129
MLLM-non-similar pairs, within class	357,383	10,829.8	5,384.0	22,248.7	154	128,893
MLLM-non-similar pairs, cross class	44,253	1,341.0	707.0	1,816.7	26	9,083
Type 1 triplets	1,638,246	49,643.8	11,337.0	150,891.8	146	862,117
Type 2 triplets	1,539,176	46,641.7	10,896.0	139,927.5	158	798,860
Type 3 triplets	1,638,246	49,643.8	11,337.0	150,891.8	146	862,117

Table 3: US design patent classes used for class-based citation analysis. Classes D31 and D33 do not exist in the USPTO design classification system and are therefore absent from the analysis.

Code	Design class	Code	Design class	Code	Design class	Code	Design class
D1	Edible products	D10	Measuring and testing devices	D19	Office supplies and equipment	D28	Pharmaceuticals and cosmetics
D2	Apparel and haberdashery	D11	Jewelry and symbolic insignia	D20	Sales, advertising, and signs	D29	Outdoor and garden equipment
D3	Travel goods and personal items	D12	Transportation	D21	Amusement devices	D30	Animal husbandry
D4	Brushware	D13	Production and distribution equipment	D22	Arms and pyrotechnics	D32	Washing and cleaning equipment
D5	Textile and fabric articles	D14	Recording, communication, and information equipment	D23	Environmental heating and cooling	D34	Material and article handling
D6	Furnishings	D15	Machines	D24	Medical and laboratory equipment	D99	Miscellaneous
D7	Food-preparation equipment	D16	Photography and optics	D25	Building units and construction		
D8	Tools and hardware	D17	Musical instruments	D26	Lighting		
D9	Packages and containers	D18	Printing and office machinery	D27	Tobacco and smoking supplies		

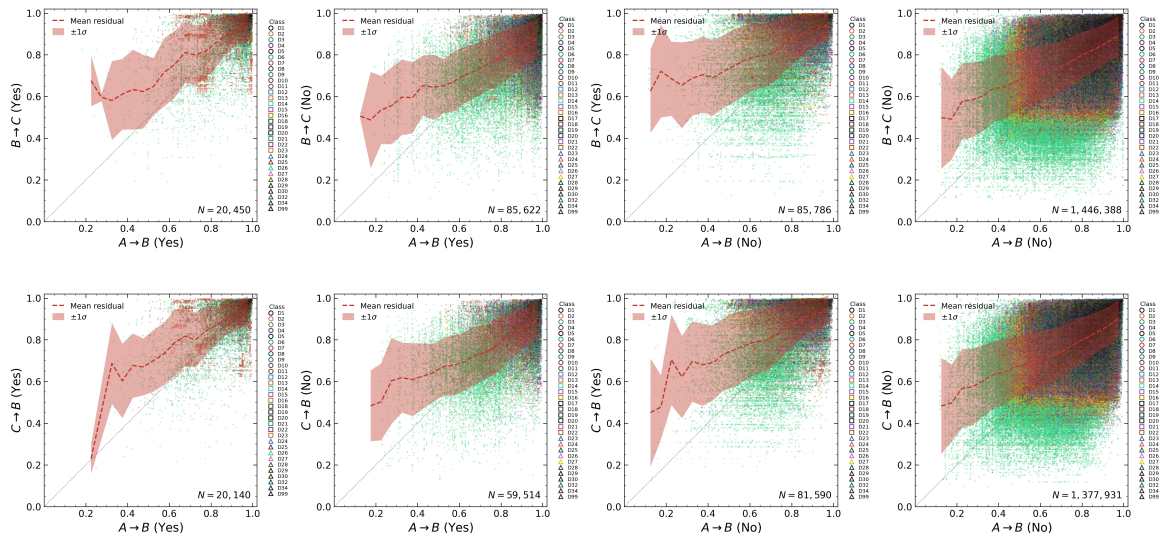


Figure 6: Scatter-band variants for Type 1 triplets (top row) and Type 2 triplets (bottom row), grouped by judgment regimes from left to right: (Yes, Yes), (Yes, No), (No, Yes), and (No, No).

Table 4: **Asymmetric judgment regimes produce opposite-sign mean residuals ($\bar{r}$) for both Type 1 and Type 2 triplets.** Each row corresponds to one of the eight MLLM judgment regimes (triplet type $\times$ label pair). $\bar{r}$ is the mean of $\text{sim}_{BC/CB} - \text{sim}_{AB}$; positive values indicate that the relevant adjacent edge has higher cosine similarity than the AB edge on average. Zero crossing denotes the value of $\text{sim}(A, B)$ at which the bin-wise mean residual changes sign (band-plot bins, width = 0.05). The final column reports the number of triplets above the zero-crossing threshold with positive residuals.

Triplet Type	Judgment Regime	Number of Triplets	Overall Mean Residual	Positive Residual (%)	Zero Crossing of Mean Residual	Triples Above Threshold
1	(Yes, Yes)	20,450	0.0004	49.966	0.911	9,621
1	(Yes, No)	85,622	-0.0508	30.216	0.775	57,320
1	(No, Yes)	85,786	0.0518	69.826	0.923	15,931
1	(No, No)	1,446,388	0.0009	50.216	0.796	603,643
2	(Yes, Yes)	20,140	0.0113	55.804	0.912	9,472
2	(Yes, No)	59,514	-0.0305	36.682	0.761	40,090
2	(No, Yes)	81,590	0.0566	72.205	0.925	14,012
2	(No, No)	1,377,931	0.0036	51.335	0.798	573,848

Table 5: **Same-company pairs have substantially higher A–C cosine similarity than connected different-company or random pairs;** connected different-company and random pairs are nearly indistinguishable ($d < 0.08$, $\text{AUC} \approx 0.52$). Random group is class-distribution-matched to the Diff. group. Cohen’s d and AUC-ROC are one-sided (group $>$ Random).

Type	Group	N	$\text{sim}(A, C)$					vs. Random	
			Mean	Median	Q25	Q75	SD	d	AUC
Type 1	Same company	175,273	0.873	0.901	0.828	0.945	0.098	0.706	0.719
Type 1	Diff. company, conn.	512,832	0.799	0.822	0.729	0.893	0.122	0.071	0.523
Type 1	Random (matched)	512,832	0.790	0.812	0.718	0.884	0.123	—	—
Type 2	Same company	207,443	0.883	0.913	0.841	0.953	0.098	0.803	0.747
Type 2	Diff. company, conn.	486,149	0.798	0.821	0.728	0.892	0.122	0.073	0.523
Type 2	Random (matched)	486,149	0.789	0.812	0.716	0.884	0.123	—	—

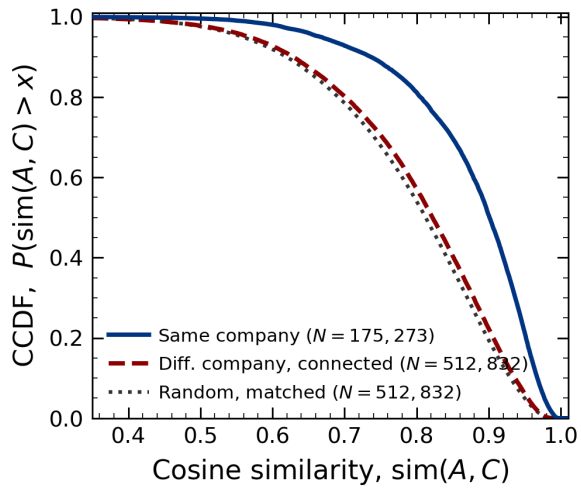


Figure 7: **Type 1 citation chains show the same qualitative pattern as Fig. 5.** Same-company A–C pairs have higher similarity, while connected different-company pairs closely track class-matched random pairs.